

Shipping Efficiency and Route Optimization Analysis for Nassau Candy Distributor

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Abstract— Efficient logistics plays a vital role in maintaining customer satisfaction and reducing operational costs in distribution businesses. This study evaluates the shipping performance of Nassau Candy Distributor using historical shipment records containing order dates, ship dates, customer locations, shipping modes, and factory information. After preprocessing and feature engineering, shipping lead time, route efficiency, route volume, and geographic performance were analyzed. The study also identified significant data quality issues caused by inconsistent shipment dates, which affected the interpretation of delivery performance. Despite these limitations, the analysis highlights key logistics bottlenecks and provides recommendations for improving routing strategies, data governance, and operational efficiency.

Keywords— Shipping Analytics, Route Optimization, Logistics, Business Intelligence, Supply Chain

I. INTRODUCTION

Logistics performance is a critical factor influencing operational efficiency, customer satisfaction, and overall profitability in distribution businesses. Nassau Candy Distributor ships products from multiple manufacturing locations to customers across the United States, making transportation efficiency an important aspect of business operations. This research evaluates shipping performance by examining delivery lead times, shipping routes, geographic distribution, and transportation modes. The objective is to identify operational bottlenecks and provide insights that support data-driven logistics optimization.

II. DATASET AND METHODOLOGY

The analysis was conducted using more than 10,000 shipment records containing order dates, shipment dates, customer locations, sales information, product details, and shipping modes. Factory locations were derived through product-to-factory mapping to construct shipping routes between manufacturing facilities and destination states. Data preprocessing included date conversion, removal of invalid records, lead-time calculation, geographic standardization, and feature engineering. Additional metrics such as Route Efficiency Score, Delay Indicator, and Route Volume were generated to evaluate logistics performance.

III. RESULTS AND DISCUSSION

The analysis revealed considerable variation in shipping performance across different routes and regions. Several high-volume routes experienced longer delivery times, indicating potential operational bottlenecks that require attention. Geographic analysis showed that certain destination states consistently recorded higher average lead times than others,

suggesting opportunities for regional optimization. Furthermore, comparisons between shipping modes demonstrated noticeable differences in delivery performance; however, interpretation of these results was limited by a significant data quality issue. Numerous shipment records contained ship dates from the year 2026 while their corresponding order dates belonged to 2024, producing unrealistic lead times exceeding 900 days. Rather than modifying the source data, these anomalies were retained and documented for stakeholder validation, ensuring analytical transparency.

IV. RECOMMENDATIONS

To improve logistics performance, the organization should first validate shipment dates within the source systems and introduce automated data validation rules during ETL processing. High-volume routes experiencing persistent delays should be prioritized for operational review, while regional bottlenecks should be investigated to improve warehouse allocation and transportation planning. Regular monitoring of route efficiency and shipping performance will enable earlier detection of operational issues and support continuous process improvement.

V. CONCLUSION

This study demonstrates that shipping analytics provides valuable operational insights for distribution organizations while emphasizing the importance of reliable data quality. Although meaningful trends regarding route efficiency and regional performance were identified, the presence of inconsistent shipment dates significantly affected lead-time analysis. Addressing these data quality issues will improve the accuracy of future analyses and enable more effective logistics

planning, ultimately reducing delivery delays and enhancing customer satisfaction.

REFERENCES

- [1] Nassau Candy Distributor Shipping Dataset (2024–2025).
- [2] Python Documentation
- [3] Streamlit Documentation